

Fernando Román Hidalgo

ROBOTICS AND AUTOMATION ENGINEER

41909, Salteras, Sevilla, Spain

☎ 611467316 ✉ ferromhid@gmail.com 🗣 FerXxk 🌐 Fernando Román Hidalgo 🏠 Portfolio



Summary

Engineer with a **multidisciplinary profile and systemic vision**, featuring professional experience in developing technological solutions and currently completing postgraduate studies. I possess a **high capacity for abstraction in solving complex problems** and **accelerated self-taught learning**, enabling me to provide significant value in technically demanding projects. My objective is to consolidate an ambitious career plan, with a focus on automation and embedded software, within dynamic and innovative environments where continuous improvement is the standard.

Experience

Software Engineer

🇦🇪 Artificial Aerospace & Defense | Seville, Spain

FULLSTACK DEVELOPMENT, AUTOMATION AND PROJECT MANAGEMENT

April 2025 - Present

• Management and Client Relations:

- Project management for the development of a process automation application for a major aerospace company.
- Direct customer interaction for requirements definition and specification.
- Technical leadership and coordination in the development of custom solutions.

• Technical Development and Automation:

- Fullstack development, applying coding best practices, efficiency, and organization.
- Design and implementation of a digital twin for a composite material production plant, oriented towards simulation and optimization of industrial processes.
- Software integration and deployment using Docker containers.
- Automation and orchestration of workflows with Apache Airflow.
- Development of AI models for early failure detection
- Complex production scheduling algorithms using Google OR-Tools' CP-SAT solver, implementing variable capacity-load constraints and custom priority weighting systems.
- Acquisition and interpretation of data from industrial sensors.

Projects

To see all my projects in detail, check out my **Interactive Portfolio**.

Honors and Awards

Winner

National Competition Robotics League (LNRC) | Spain

MINISUMO CATEGORY

2024

Education

Master's Degree in Electronic Engineering, Robotics and Automation	 University of Seville Seville, Spain 2025 - 2026
Bachelor's Degree in Electronic Engineering, Robotics and Mechatronics GPA: 7.5/10	 University of Seville Seville, Spain 2021 - 2025
Technological Baccalaureate (Graduated with Honors distinction)	 IES Heliche Seville, Spain 2019 - 2021

Certifications

Advanced Python Course ONLINE	 Idexa Jan 2026
Certificate in Advanced English (C1 Advanced)	 Cambridge Assessment English Sep 2024
Basic Life Support and Automated External Defibrillation	 Ympetu 2020
Aquatic Lifeguard	 Ympetu 2020

Skills

Programming	C, C++, Python (OpenCV), MATLAB, Simulink, ROS/ROS2, JavaScript, React, VBA, HTML, CSS, SQL
Electronics	Digital/Analog electronics, microcontrollers (MSP430, ESP32, Arduino...), RTOS, FPGA, automation and control
Simulation	System modeling and simulation (Gazebo, Webots), PID control design, embedded systems integration
DevOps	Docker, Linux, Git, CI/CD, Apache Airflow
Protocols	TCP/IP, MQTT, Zigbee, CAN bus, UART, I2C, SPI
3D Design	Fusion 360, FreeCAD

Soft Skills

Capabilities	Communication, leadership, adaptation, proactivity, critical thinking, punctuality, autonomy, management and teamwork
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Languages

Spanish	Native
English	Professional competence (C1)
French	Basic competence